

STAT 240 Lab 9/Sample Final Coding Exam

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Please read all instructions on this page before beginning the exam.

1. This exam will start 5 minutes after the start of your assigned lab. You will have 45 minutes to complete the exam, **which includes the time required to submit to crowdmark**.
2. Any work not submitted by the end of the lab **you are registered for** will receive 0.
3. You can use the slides from the class and any notes you have created. If you have a web browser open, it can only be used to access crowdmark, create your pdf file, or to view course materials.
4. You must submit a **rendered** pdf file for your exam to be graded. You should use the template file provided last week, which contains all packages needed. **Any submission not in this format will receive 0.**

Dataset

For this exam we will make use of the following dataset, `ad_data`, which you can download from the internet using the following command. This will save a `csv` file locally, in your current working directory.

```
download.file("https://owenward.github.io/files/ad_data.csv", "ad_data.csv")
```

We will only use this file for Q4.

Problem 1

In this question we will use the `iris` dataset (included in R) to perform k-means clustering. The following code should be included to first process the data (as we will only use 2 predictors).

```
data(iris)
iris_data <- iris[, c(1, 2)]
```

Problem 1a

Using `iris_data`, perform k-means clustering with $k = 3$ clusters in the data. Be sure to set the maximum number of iterations to 100 and perform at least 10 random starts. Save the estimated clusters as `iris_kmeans`.

(3 points)

Problem 1b

Plot the data in `iris_data` as a scatter plot, where the points are colored by the estimated clusters you obtain from `iris_kmeans`.

(2 points)

Problem 2

The `stringr` package contains a dataset `sentences` consisting of some short sentences. We will use a subset of these words, which can be obtained with the following code.

```
library(stringr)
small_sentences <- sentences[1:10]
words_vector <- unlist(str_split(small_sentences, "\\s+"))[1:20]
```

Problem 2a

Using `words_vector`, construct a regular expression to extract the first 4 letter word from each of these sentences. You can show a vector where the element is either a 4 letter words in `words_vector` or NA if the corresponding element in `words_vector` is not a 4 letter word.

(2 points)

Problem 2b

Again using `words_vector`, return all words which start with the letter `b` or `c`.

(2 points)

Problem 2c

Using `words_vector`, extract a vector containing only words which have additional characters beyond the 26 letters in the alphabet.

(1 points)

Problem 3

Suppose you were given a statistical model which had the following log-likelihood function,

$$\log \mathcal{L} = -n\theta + \log(\theta) \sum_{i=1}^n x_i,$$

where θ is the parameter in this model, n is the number of observations and each of these observations are $x_1, x_2, \dots, x_n$.

Here `log` is the natural logarithm.

Problem 3a

Write a function `pois_llh` which will take in as input values of `theta`, `n`, and a vector `x` and compute the log-likelihood. What does this return for the values `pois_llh(theta = 1, n = 5, x = c(1, 2, 3, 4, 5))`?

(3 points)

Problem 3b

With `n` and `x` fixed at the values in the previous part, evaluate this function at the range of values of `theta_vals = seq(from = 0.1, to = 8, length.out = 100)`. Construct a plot of these values of the log-likelihood and identify the maximum likelihood estimate using the plot. You do not have to identify it exactly, just an educated guess based on the plot.

(2 points)

Problem 4

In this problem we will use KNN to do classification, predicting a binary variable `purchased` in `ad_data.csv` (download instructions are given on the first page).

The following code will split this data into training and test data, assuming you have read it in as `ad_data`.

```
set.seed(100)
prop_train <- 0.7
prop_test <- 1 - prop_train

n_data <- nrow(ad_data)

train_test <- sample(c("Train", "Test"),
                    size = n_data, replace = TRUE,
                    prob = c(prop_train, prop_test))

training_data <- ad_data[train_test == "Train", ]
testing_data <- ad_data[train_test == "Test", ]
```

Problem 4a

Fit KNN on this training data where we use the variable `EstimatedSalary` as the single predictor, with $K = 5$ and store the fit in an object called `fit_knn`. To do this you also need to pass in the predictors in `testing_data`. You will need to use the `class` package. (3 points)

Problem 4b

Compute the **Accuracy** of this KNN model on the test data. (2 points)